

Sarah Boyd

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2/25/25

Three phase power

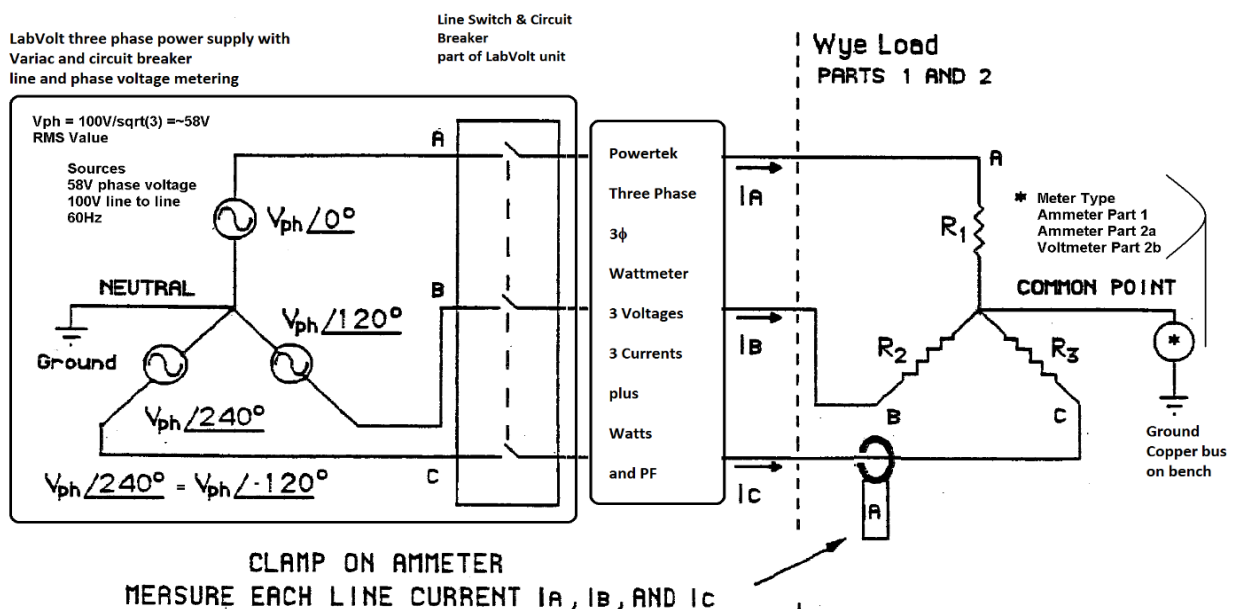
Introduction:

This experiment examines the behavior of three-phase (3Ø) power in both balanced and unbalanced Wye and Delta resistive load configurations. A 100 VRMS line-to-neutral voltage (173 VRMS line-to-line) was supplied. To prevent excessive heating of the resistors, power was applied only briefly while measurements were taken. Voltage and current relationships were analyzed to understand how they differ between Wye and Delta power circuits, as well as balanced and unbalanced three phase power circuits. For balanced sources for three phase electrical power sources, the balanced can refer to the sources having the same input voltage and the angles summing up to $360/3 = 120^\circ$ (for example, $v_a = 100\angle 0^\circ$, $v_b = 100\angle -120^\circ$, $v_c = 100\angle 120^\circ$). The objective of this lab was to explore and understand three-phase power and verify measured results against calculated data, which had the added benefit of understanding the calculation process of dealing with three phase power circuits. **The most important takeaway** from this lab was understanding the process for calculating balanced wye

and delta, as this information is particularly relevant. A step by step guide created using the calculations from this lab can be found after the conclusion section.

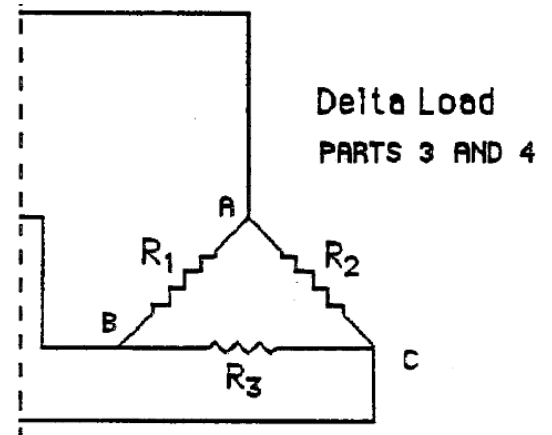
EQUIPMENT:

1. 3 Resistor breadboard, 100/200
2. LabVolt 3 power supply unit
3. Clamp-on Ammeter preferred or bench ammeter
4. Handheld Voltmeter (user supplied),
5. Powertek Three Phase 3 power meter, not all groups will use this unit.



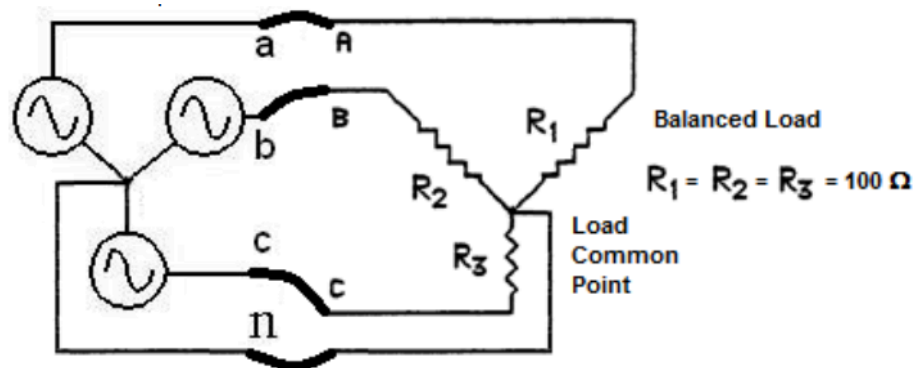
IMG 1. Overall circuit setup diagram for Wye loads, for part 1 and 2.

USE THE CLAMP ON AMMETER TO
MEASURE EACH PHASE CURRENT
 I_{AB} , I_{BC} , AND I_{CA}



IMG 2. Overall circuit setup diagram for Delta loads, for part 3 and 4.

Part 1a: Balanced Wye with connection from neutral to common



IMG 3. The circuit shown above is a balanced WYE load. Source neutral and common points connected.

Measurements

Vcp-N	VA-N	VB-N	VC-N	VC-A	IA	IB	IC	IN
0	100.2vr ms	99.7 vrms	99.0vr ms	173	0.99	0.97	1.01	0.02

Table 1. Measured data from the balanced wye load circuit.

Calculations

Vcp-N	VA-N	VB-N	VC-N	VA-B	VB-C	VC-A	IA	IB	IC	IN
0	100<0vrms	100<-120vrms	100<120vrms	173.2<30	173.2<-90	173.2<-210	1<0 Arms	1<-120Arms	1<120 Arms	0

Table 2. Calculated data from the balanced wye load circuit.

The calculations are shown below:

- 1) Calculate total impedance per line

$$Z_a = R_1 = 100\Omega < 0^\circ$$

$$Z_b = R_2 = 100\Omega < 0^\circ$$

$$Z_c = R_3 = 100\Omega < 0^\circ$$

- 2) Given load voltages:

$$V_{an} = 100 < 0 \text{ vrms}$$

$$V_{bn} = 100 < -120 \text{ vrms}$$

$$V_{cn} = 100 < 120 \text{ vrms}$$

- 3) Calculate the line currents

$$I_{aA} = \frac{V_{an}}{Z_a} = \frac{100\text{vrms} < 0}{100\Omega < 0} = 1 < 0^\circ \text{ Arms}$$

$$I_{bB} = I_{aA} < (\theta_B - |\theta_A|) = 1 < (-120 - |0|) = 1 < -120 \text{ Arms}$$

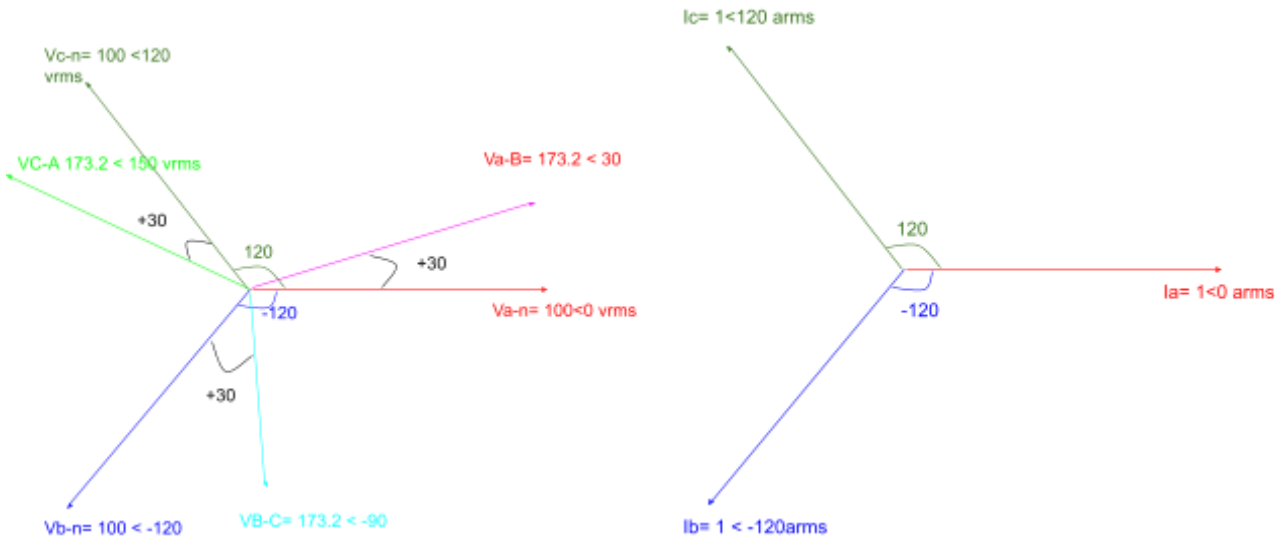
$$I_{cC} = I_{aA} < (\theta_C - |\theta_A|) = 1 < (120 - |0|) = 1 < 120 \text{ Arms}$$

4) Calculate the line voltages

$$V_{a-b} = V_a \sqrt{3} < 30 = 100\sqrt{3} < 30 = 173.2 < 30$$

$$V_{b-c} = V_{a-b} < (\theta_B + |\theta_A|), V_{a-b} < (-120 + |30|) = 100\sqrt{3} < -90$$

$$V_{c-a} = V_{a-b} < (\theta_C + |\theta_A|), V_{a-b} < (120 + 30) = 100\sqrt{3} < 150 \text{ (or } -210)$$

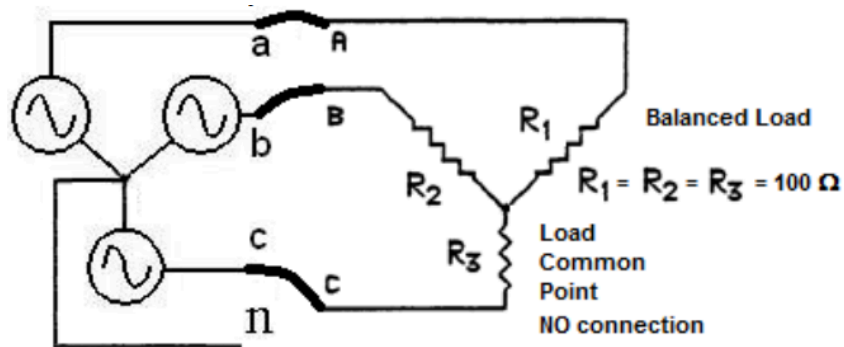


IMG 4. Phasor diagram of the voltages and currents for the balanced wye circuit.

In a balanced wye circuit with a common neutral connection such as the one we built, the sum of the three-phase currents equals zero when treated as vectors. Mathematically this can be expressed as the following: $I_a + I_b + I_c = I_n$. Each phase current is equal in magnitude but separated by 120° in phase, meaning their contributions cancel out when summed: $120^\circ + -120^\circ + 0^\circ = 0^\circ$. This equation

implies that no current flows through the neutral conductor in an ideal balanced system. This means that the neutral connected to LCP is removable. The phrase " $1 + 1 + 1 = 0$ " represents this concept and how the phase relationships result in a net sum of zero.

Part 1b: Balanced Wye load without common point to neutral



IMG 5. *Balanced Wye load without neutral and common point connected. This circuit is similar to the circuit in part 1a, except that the neutral line is removed.*

Measurements

V _{cp-N}	V _{A-N}	V _{B-N}	V _{C-N}	V _{C-A}	I _A	I _B	I _C
1.4	100.2	100.7	98.2	173	0.99	0.97	1.0

Table 3. *Measured data from the balanced wye without neutral circuit.*

Calculations

Vcp-N	VA-N	VB-N	VC-N	VA-B	VB-C	VC-A	IA	IB	IC
0	100<0 vrms	100<-120vrms	100<120vrms	173.2<30 vrms	173.2<-90 vrms	173.2<-210	1<0 Arms	1<-120 Arms	1<120 Arms

Table 4. Calculated data from the balanced wye without neutral circuit.

1) Calculate total impedance per line

$$Z_a = R_1 = 100\Omega < 0^\circ$$

$$Z_b = R_2 = 100\Omega < 0^\circ$$

$$Z_c = R_3 = 100\Omega < 0^\circ$$

2) Given load voltages:

$$V_{an} = 100 < 0 \text{ vrms}$$

$$V_{bn} = 100 < -120 \text{ vrms}$$

$$V_{cn} = 100 < 120 \text{ vrms}$$

$$* V_{cpn} = 0 < 0$$

3) Calculate the line voltages

$$V_{a-b} = V_{an}\sqrt{3} < 30, \quad 100\sqrt{3} < 30 = 173.2 < 30$$

$$V_{b-c} = V_{bn}\sqrt{3} < -90, \quad V_{bn}\sqrt{3} < (\theta_B + |\theta_A|), \quad 100\sqrt{3} < (-120 + |30|) = 173.2 < -90$$

$$V_{c-a} = V_{cn}\sqrt{3} < 150, \quad V_{cn}\sqrt{3} < (\theta_C + |\theta_A|), \quad 100\sqrt{3} < (120 + 30) = 173.2 < 150$$

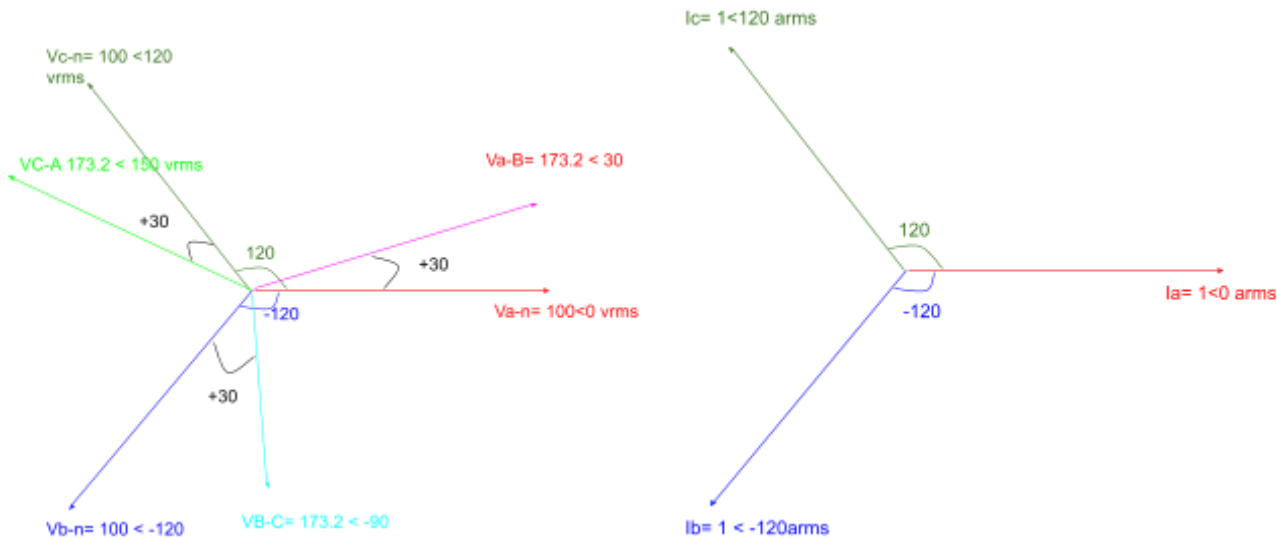
4) Calculate the line currents

$$I_{aA} = \frac{V_{an}}{Z_a} = \frac{100\text{vrms} \angle 0}{100\Omega \angle 0} = 1 \angle 0^\circ \text{ Arms}$$

$$I_{bB} = I_{aA} \angle (\theta_B - |\theta_A|) = 1 \angle (-120^\circ - |0|) = 1 \angle -120^\circ \text{ Arms}$$

$$I_{cC} = I_{aA} \angle (\theta_C - |\theta_A|) = 1 \angle (120^\circ - |0|) = 1 \angle 120^\circ \text{ Arms}$$

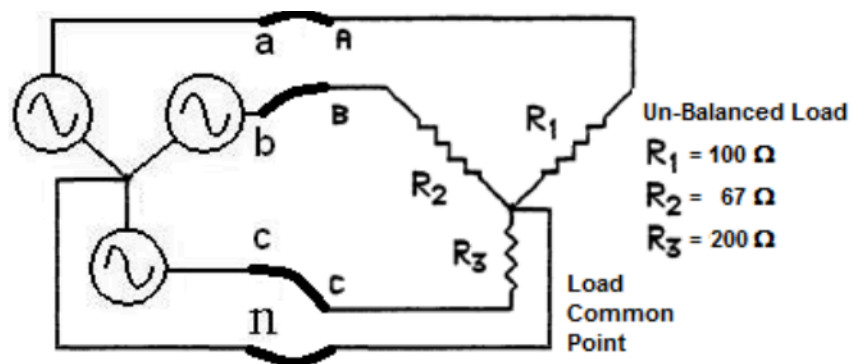
As shown by the calculations and measurements, the results of removing the connection between neutral and common yields almost exactly the same result. This is why the data in part 1a and part 1b is so similar. V_{cp-N} is also confirmed as zero in the calculations, and close to zero in the measurements. As in part 1a, each phase is equal in magnitude but separated by 120° in phase, meaning their contributions cancel out when summed: $120^\circ + -120^\circ + 0^\circ = 0^\circ$. So the formula $I_a + I_b + I_c = I_n$ remains true even when removing the neutral to common connection.



IMG 6. Phasor diagram of the voltages and currents

Part 2a: Unbalanced Wye load with common point to neutral connection

The circuit is rewired and connection wires added to the resistor breadboard to now provide $100\ \Omega$, $67\ \Omega$ (i.e. $100\ \Omega$ parallel $200\ \Omega$) and $200\ \Omega$ as the load resistors. These resistors are connected as R_1 , R_2 and R_3 (in order) to provide an unbalanced Wye configuration. Then, the data is measured and calculated like the process in part 1.



IMG 7. Unbalanced Wye load with common point and neutral connected.

Measurements

Vcp-N	VA-N	VB-N	VC-N	IA	IB	IC	IN
0vrms	100.7vrms	99.4vrms	100.2vrms	1.00	1.43	0.44	0.84

Table 5. Measured data from the unbalanced wye circuit with neutral and common connected.

Calculations

Vcp-N	VA-N	VB-N	VC-N	VC-A	IA	IB	IC	IN
0vrms	100<0v rms	99.4<-1 20vrms	100.2< 120vrm s	173.v< 150 vrms	1.01ar ms	1.48arn s	0.50ar ms	0.86= arms

Table 6. Calculations from the unbalanced wye circuit with neutral and common connected.

The work for the calculations are below:

- 1) Calculate total impedance per line

$$Z_a = R_1 = 100\Omega < 0^\circ$$

$$Z_b = R_2 = 67\Omega < 0^\circ$$

$$Z_c = R_3 = 200\Omega < 0^\circ$$

- 2) Load voltages in polar form:

$$V_{an} = 100.7 < 0 \text{ vrms}$$

$$V_{bn} = 99.4 < -120 \text{ vrms}$$

$$V_{cn} = 100.2 < 120 \text{ vrms}$$

- 3) Calculated line currents

$$I_A = \frac{V_{an}}{Z_a} = \frac{100.7 \angle 0^\circ}{100\Omega \angle 0^\circ} = 1.007 \angle 0^\circ \text{ Arms in polar form}$$

Converted to rectangular form: $I_A = 1.0, 0j$

$$I_B = \frac{V_{bn}}{Z_b} = \frac{99.4 \angle -120^\circ}{67 \angle 0^\circ} = 1.483 \angle -120^\circ \text{ Arms in polar form}$$

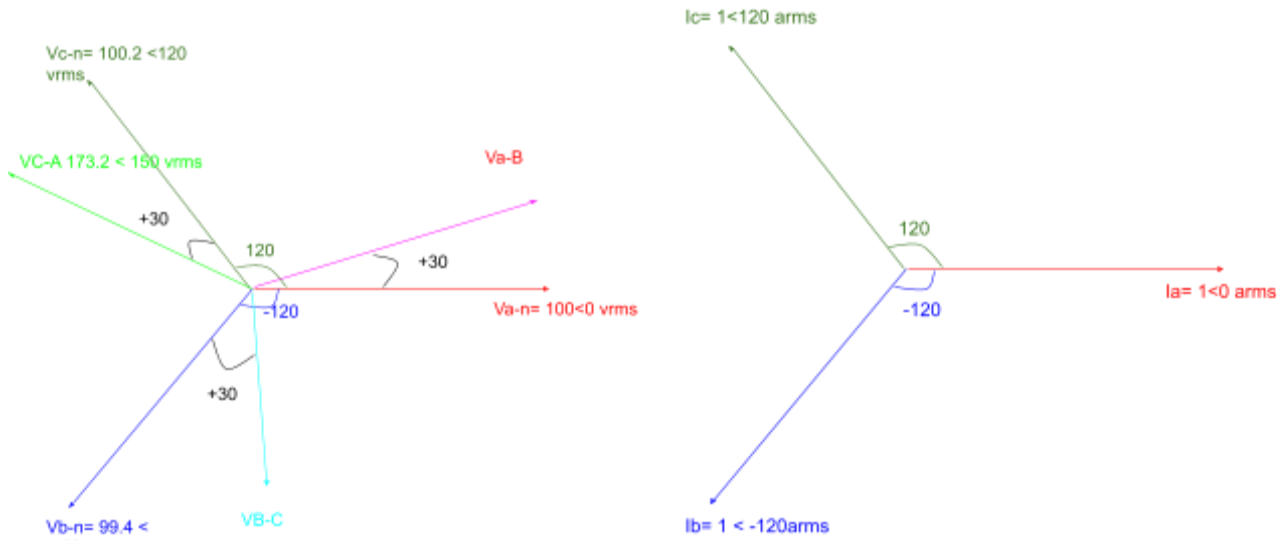
Converted to rectangular form: $I_B = -0.72, -1.24j$

$$I_C = \frac{V_{cn}}{Z_c} = \frac{100.2 \angle 120^\circ}{200 \angle 0^\circ} = 0.501 \angle 120^\circ \text{ Arms in polar form}$$

Converted to rectangular form: $I_C = -0.22, 0.38j$

$$I_N = |I_A + I_B + I_C| = \sqrt{I_A^2 + I_B^2 + I_C^2} = \sqrt{(0.065)^2 + (-0.86)^2} = 0.86 \text{ arms}$$

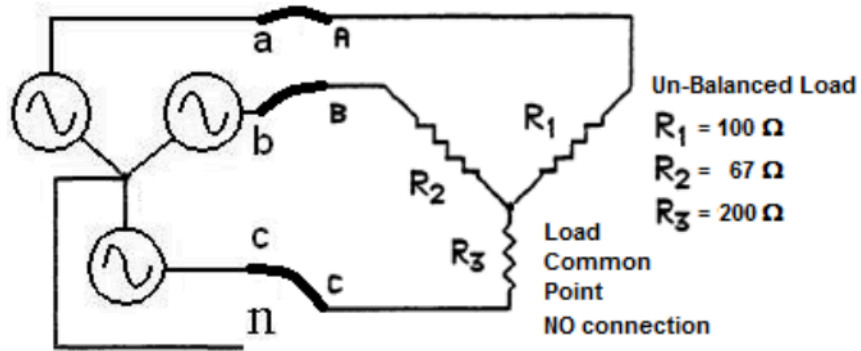
One expects the neutral current to be zero or very close to zero for the balanced Wye configuration. However, when the resistance in each phase is unbalanced, the neutral current will not be zero. In this case, the neutral current can be calculated by applying Kirchhoff's current law at the common point/node i.e. CP of the load. When the load CP is connected to neutral of the source this Neutral current is the sum of line/phase currents.. Vector addition using rectangular components (real and imaginary) or graphically with vectors (magnitude and phase) is needed. When the CP is not connected to the neutral (i.e. floating), the total current from the three phases must be zero and in this case the CP voltage (magnitude and phase angle) will adjust accordingly for the imbalance. This CP to neutral voltage, V_{cp-N} , can be determined by again applying Kirchhoff's current law at the CP assuming that V_{cp-N} or simply V_{cp} has an unknown magnitude and phase. In both cases, the phase voltages of the source with respect to neutral remain unchanged.



IMG 8. *Phasor diagram of the voltages and currents*

Part 2b: Unbalanced Wye load with *no* common point to neutral connection

The common point to neutral connection, present in part 2a, is now disconnected. The data is now measured again. Now the neutral current is zero as there is no neutral wire. Instead of measuring the neutral current, we eliminated the neutral and measured voltage between the common point and neutral using the bench voltmeter. A voltage around 30 volts appeared between the neutral and the common point.



IMG 9. Circuit 2b- unbalanced WYE load with the neutral and common point not connected.

Measurements

Vcp-N	VA-N	VB-N	VC-N	IA	IB	IC	IN
30.0	103.4	76.6	127.5	1.02	1.10	0.56	0

Table 9. Measurements from the unbalanced wye circuit with neutral and common not connected.

Calculations

Vcp-N	VA-N	VB-N	VC-N	IA	IB	IC	IN
30.0vrms	103< 0 vrms	77 < -120 vrms	129 < 120vrms	1.00 <0 arms	1.49 < -120 arms	0.50 < 120 arms	0.86arm s

Table 10. Calculations from the unbalanced wye circuit with neutral and common not connected.

Calculations are shown below:

1) Calculate total impedance per line

$$Z_a = R_1 = 100\Omega < 0^\circ$$

$$Z_b = R_2 = 67\Omega < 0^\circ$$

$$Z_c = R_3 = 200\Omega < 0^\circ$$

2) Given load voltages:

$$V_{an} = 100 < 0 \text{ } vrms$$

$$V_{bn} = 100 < -120 \text{ } vrms$$

$$V_{cn} = 100 < 120 \text{ } vrms$$

$$* V_{cpn} = 0 < 0$$

3) Calculate the line voltages

$$V_{a-b} = V_{an}\sqrt{3} < 30, \quad 100\sqrt{3} < 30 = 173.2 < 30$$

$$V_{b-c} = V_{bn}\sqrt{3} < -90, \quad V_{bn}\sqrt{3} < (\theta_B + |\theta_A|), \quad 100\sqrt{3} < (-120 + |30|) = 173.2 < -$$

$$V_{c-a} = V_{cn}\sqrt{3} < 150, \quad V_{cn}\sqrt{3} < (\theta_C + |\theta_A|), \quad 100\sqrt{3} < (120 + 30) = 173.2 < 150$$

4) Calculate the line currents

$$I_A = \frac{V_{an}}{Z_a} = \frac{100\angle 0^\circ}{100\angle 0^\circ} = 1.007 < 0^\circ \text{ Arms in polar form}$$

$$\text{Converted to rectangular form: } I_A = 1.0, 0j$$

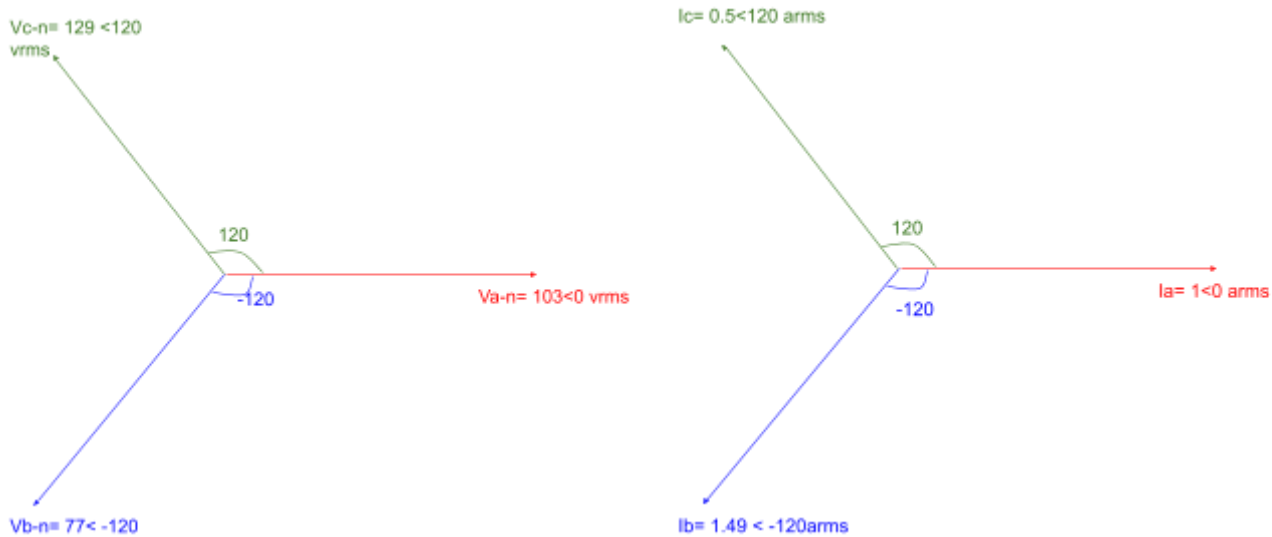
$$I_B = \frac{V_{bn}}{Z_b} = \frac{99.4 \angle -120}{67 \angle 0} = 1.49 \angle -120^\circ \text{ Arms in polar form}$$

Converted to rectangular form: $I_B = -0.72, -1.24j$

$$I_C = \frac{V_{cn}}{Z_c} = \frac{100.2 \angle 120}{200 \angle 0} = 0.5 \angle 120^\circ \text{ Arms in polar form}$$

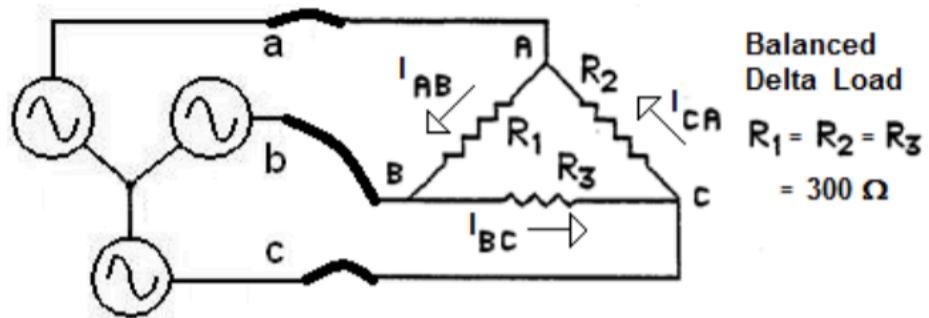
Converted to rectangular form: $I_C = -0.22, 0.38j$

$$I_N = |I_A + I_B + I_C| = \sqrt{I_A^2 + I_B^2 + I_C^2} = \sqrt{(0.065)^2 + (-0.86)^2} = 0.86 \text{ arms}$$



IMG 10. Phasor diagram of the voltages and currents

Part 3: Balanced Delta with line to neutral connection



IMG 8. Circuit 3- balanced delta with line to neutral connection.

Measured

VA-B	VB-C	VC-A	IA-B	IB-C	IC-A	IA	IB	IC
175.1	173.9	175.1	0.53	0.52	0.53	0.98	0.96	0.96

Table 11. Measurements from the balanced delta circuit with neutral and common connected.

Calculated

VA-B	VB-C	VC-A	IA-B	IB-C	IC-A	IA	IB	IC
173.2 <30	173.2 -90	< 173.2 50	$\frac{3}{\sqrt{3}}$ <30	$\frac{3}{\sqrt{3}}$ <-90	$\frac{3}{\sqrt{3}}$ <150	1<0 arms	1<-12 0 arms	1<120 arms

Table 12. Calculations from the balanced delta circuit with neutral and common connected.

The calculations are shown below:

1) Calculate total impedance per line

$$Z_a = R_1 = 300\Omega < 0^\circ$$

$$Z_b = R_2 = 300\Omega < 0^\circ$$

$$Z_c = R_3 = 300\Omega < 0^\circ$$

2) Given load voltages:

$$V_{an} = 100 < 0 \text{ } vrms$$

$$V_{bn} = 100 < -120 \text{ } vrms$$

$$V_{cn} = 100 < 120 \text{ } vrms$$

3) Calculate the line voltages * +30 degree displacement

$$V_{a-b} = V_{an}\sqrt{3} < 30, \quad 100\sqrt{3} < 30 = 173.2 < 30$$

$$V_{b-c} = V_{bn}\sqrt{3} < -90, \quad V_{bn}\sqrt{3} < (\theta_B + 30), \quad 100\sqrt{3} < (-120 + 30) = 173.2 < -90$$

$$V_{c-a} = V_{cn}\sqrt{3} < 150, \quad V_{cn}\sqrt{3} < (\theta_C + 30), \quad 100\sqrt{3} < (120 + 30) = 173.2 < 150$$

4) Calculate the line currents

$$I_{AB} = \frac{V_{AB}}{Z_a} = \frac{100\sqrt{3}<30}{300\Omega<0} = \frac{\sqrt{3}}{3} < 30^\circ$$

$$I_{BC} = \frac{V_{BC}}{Z_b} = \frac{100\sqrt{3}<-90}{300\Omega<0} = \frac{\sqrt{3}}{3} < -90^\circ$$

$$I_{CA} = \frac{V_{CA}}{Z_c} = \frac{100\sqrt{3}<150}{300\Omega<0} = \frac{\sqrt{3}}{3} < 150^\circ$$

Looking at point A...

$$I_A + I_{CA} = I_{AB}, \text{ so } I_A = I_{AB} - I_{CA}, \text{ or } I_A = \left(\frac{\sqrt{3}}{3} \angle 30^\circ\right) - \left(\frac{\sqrt{3}}{3} \angle 150^\circ\right), \text{ or}$$

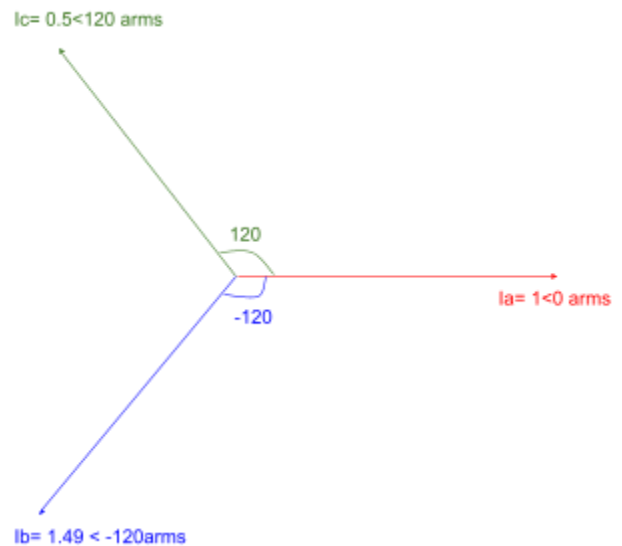
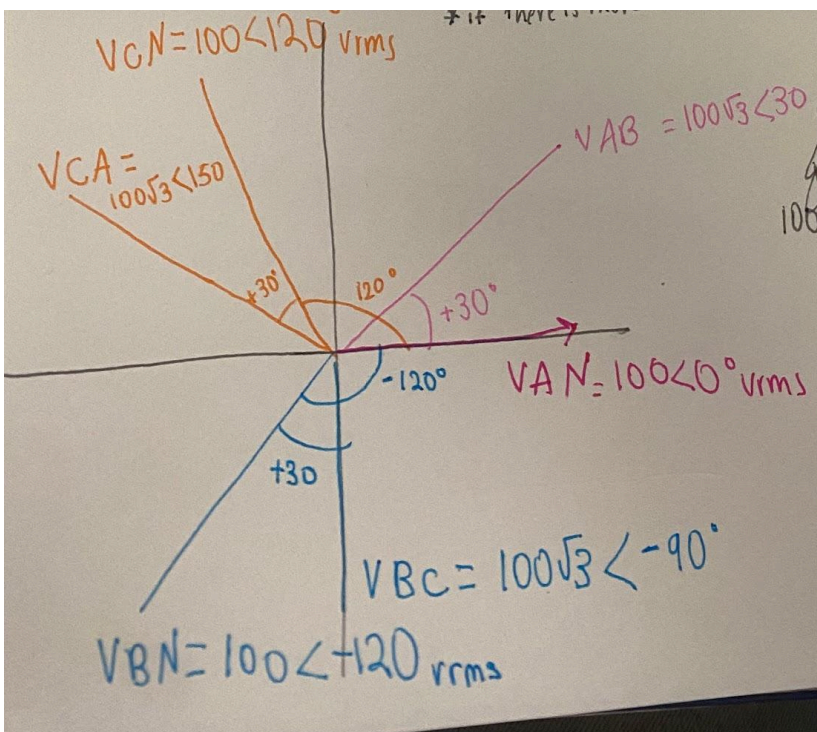
$$I_A = (0.5 - (-0.5)) \angle 0^\circ = 1 \angle 0^\circ$$

Or we can solve by...

$$I_{aA} = \frac{V_{an}}{Z_a} = \frac{100\sqrt{3} \angle 0^\circ}{100 \angle 0^\circ} = 1 \angle 0^\circ \text{ Arms}$$

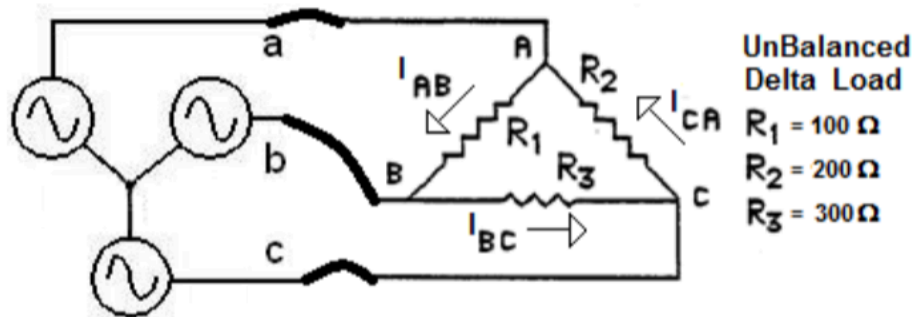
$$I_{bB} = I_{aA} \angle (\theta_B - |\theta_A|) = 1 \angle (-120^\circ - |0|) = 1 \angle -120^\circ \text{ Arms}$$

$$I_{cC} = I_{aA} \angle (\theta_C - |\theta_A|) = 1 \angle (120^\circ - |0|) = 1 \angle 120^\circ \text{ Arms}$$



IMG 9. Phasor diagram.

Part 4: Unbalanced Delta with line to neutral connection



IMG 10. Circuit 4- unbalanced delta with line to neutral connection.

Measurements

VA-B	VB-C	VC-A	IA-B	IB-C	IC-A	IA	IB	IC
173.6	172.6	171.4	0.82	0.53	1.69	2.20	1.18	2.00

Table 13. Measurements from the unbalanced wye circuit with neutral and common connected. Note that there is no neutral connection in a delta.

Calculations

VA-B	VB-C	VC-A	IA-B	IB-C	IC-A	IA	IB	IC
173.2 <30 vrms	173.2 < -90vrms	173.2< 150	$\frac{\sqrt{3}}{3} < 30$ arms	$\frac{\sqrt{3}}{2} < -90$ arms	$\frac{\sqrt{3}}{1} < 150$ arms	1<0 arms	1<-12 0 arms	1<12 0 arms

Table 13. Calculations from the unbalanced wye circuit with neutral and common connected.

The calculations are shown below:

5) Calculate total impedance per line

$$Z_a = R_1 = 100\Omega < 0^\circ$$

$$Z_b = R_2 = 200\Omega < 0^\circ$$

$$Z_c = R_3 = 300\Omega < 0^\circ$$

6) Given load voltages:

$$V_{an} = 100 < 0^\circ \text{ vrms}$$

$$V_{bn} = 100 < -120^\circ \text{ vrms}$$

$$V_{cn} = 100 < 120^\circ \text{ vrms}$$

7) Calculate the line voltages * +30 degree displacement

$$V_{a-b} = V_{an}\sqrt{3} < 30^\circ, \quad 100\sqrt{3} < 30^\circ = 173.2 < 30^\circ$$

$$V_{b-c} = V_{bn}\sqrt{3} < -90^\circ, \quad V_{bn}\sqrt{3} < (\theta_B + 30^\circ), \quad 100\sqrt{3} < (-120^\circ + 30^\circ) = 173.2 < -90^\circ$$

$$V_{c-a} = V_{cn}\sqrt{3} < 150^\circ, \quad V_{cn}\sqrt{3} < (\theta_C + 30^\circ), \quad 100\sqrt{3} < (120^\circ + 30^\circ) = 173.2 < 150^\circ$$

8) Calculate the line currents

$$I_{AB} = \frac{V_{AB}}{Z_a} = \frac{100\sqrt{3} < 30^\circ}{300\Omega < 0^\circ} = \frac{\sqrt{3}}{3} < 30^\circ$$

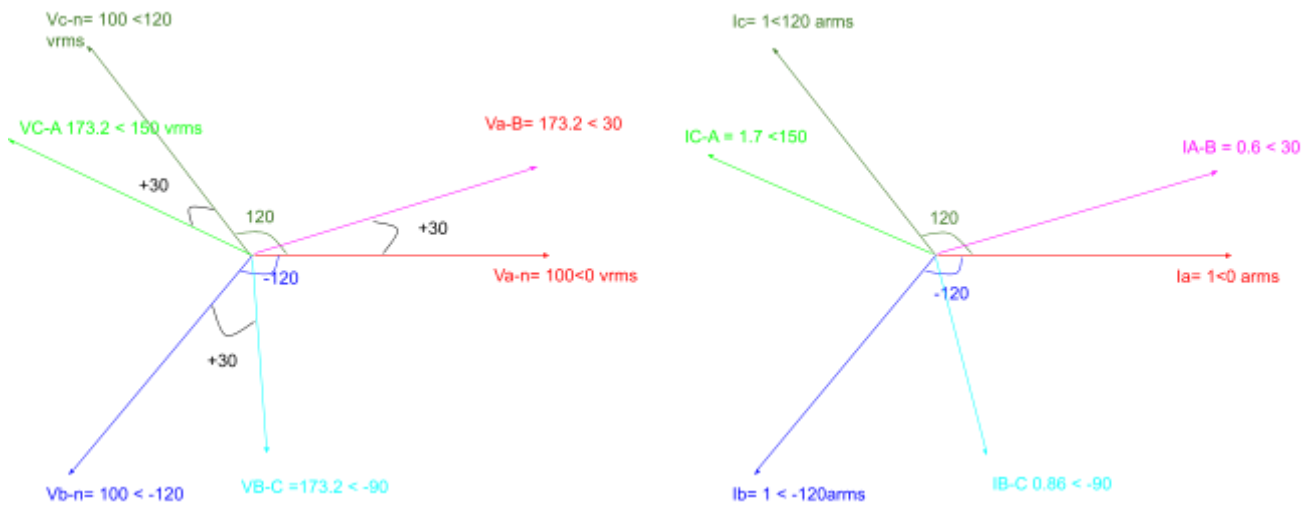
$$I_{BC} = \frac{V_{BC}}{Z_b} = \frac{100\sqrt{3} < -90^\circ}{200\Omega < 0^\circ} = \frac{\sqrt{3}}{2} < -90^\circ$$

$$I_{CA} = \frac{V_{CA}}{Z_c} = \frac{100\sqrt{3} < 150^\circ}{100\Omega < 0^\circ} = \frac{\sqrt{3}}{1} < 150^\circ$$

$$I_{aA} = \frac{V_{an}}{Z_a} = \frac{100\sqrt{3}\angle 0}{100\Omega\angle 0} = 1 \angle 0^\circ \text{ Arms}$$

$$I_{bB} = I_{aA} \angle (\theta_B - |\theta_A|) = 1 \angle (-120^\circ - |0|) = 1 \angle -120^\circ \text{ Arms}$$

$$I_{cC} = I_{aA} \angle (\theta_C - |\theta_A|) = 1 \angle (120^\circ - |0|) = 1 \angle 120^\circ \text{ Arms}$$

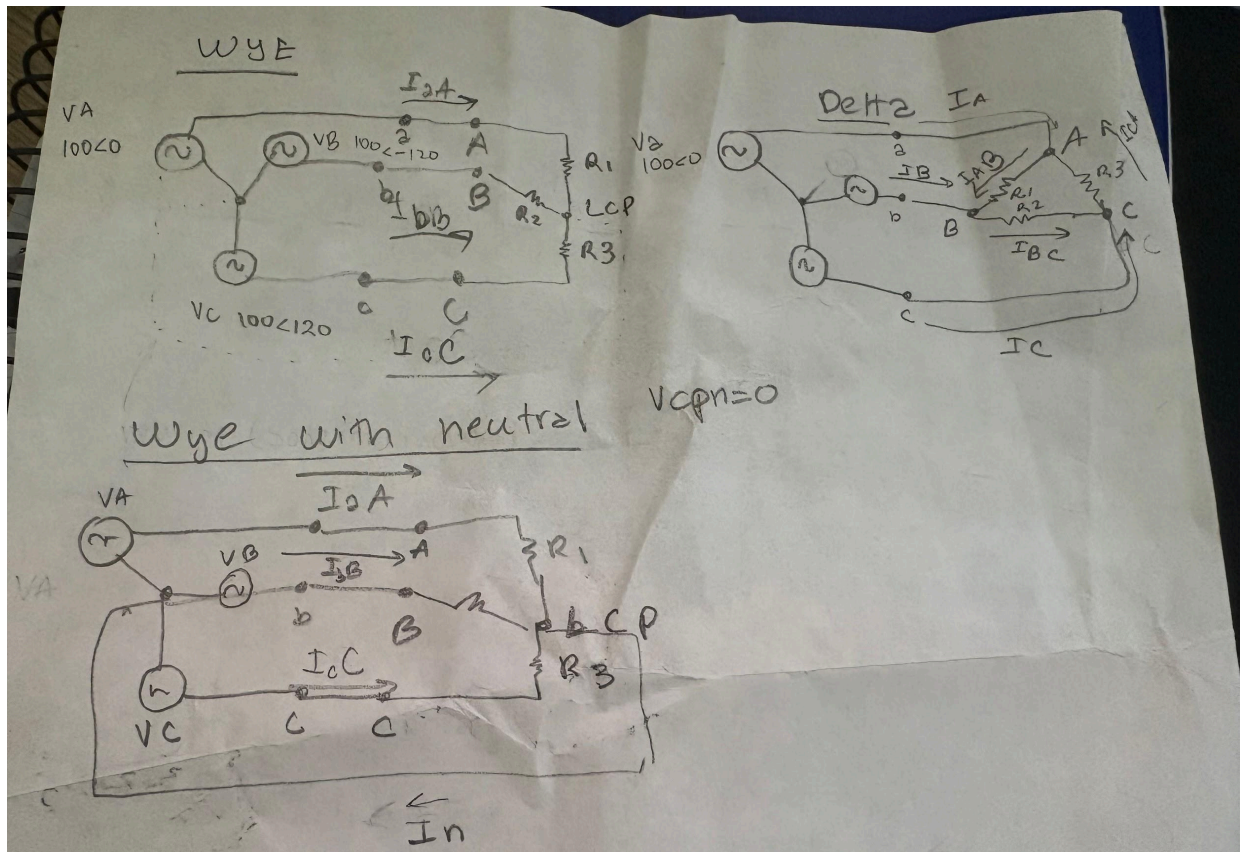


IMG 8. Phasor diagram of the voltages and currents

Conclusion:

The experiment demonstrated the behavior of three-phase power in balanced and unbalanced Wye and Delta configurations. In a balanced Wye circuit with a common neutral connection, the sum of the three-phase currents equals zero, confirming that no current flows through the neutral conductor. Removing the neutral-to-common connection yielded nearly identical results as the phase contributions canceled out. However, in an unbalanced Wye circuit, the neutral current was no longer zero and was determined by applying Kirchhoff's Current Law at the common point. When the common point was floating, the CP voltage adjusted accordingly to maintain current balance. We also noticed that $\sqrt{3}$ shows up in our calculations very often, due to trigonometry and because of 60 degree angle. The measured and calculated results were overall similar and confirmed the theory we were taught during class, and gave us practice with the calculation process for three phase power systems as well as helped us have a deeper understanding concerning how three phase power circuits work.

Process for 3 phase calculations for balanced:



IMG 9. Balanced wye (with and without neutral) and delta circuits.

Below is the process for calculating wye and delta- the process has been combined when possible in order to be more easily remembered.

- 1) Calculate total impedance per line (phase element)

$$Z_a = R_1 = 100\Omega \angle 0^\circ$$

$$Z_b = R_2 = 100\Omega \angle 0^\circ$$

$$Z_c = R_3 = 100\Omega \angle 0^\circ$$

2) Given load voltages:

$$V_{an} = v_p < \theta_{van} = 100 < 0 \text{ } vrms$$

$$V_{bn} = v_p < \theta_{vbn} = 100 < -120 \text{ } vrms$$

$$V_{cn} = v_p < \theta_{vcn} = 100 < 120 \text{ } vrms$$

$$* V_{cp-n} = 0 < 0, \text{ if applicable}$$

3) Calculate the line to line voltages

$$V_{a-b} = V_{an}\sqrt{3} < \theta_{van} + 30 = 100\sqrt{3} < 0 + 30 = 100\sqrt{3} < 30$$

$$V_{b-c} = V_{bn}\sqrt{3} < \theta_{vbn} + 30 = 100\sqrt{3} < -120 + 30 = 100\sqrt{3} < -90$$

$$V_{c-a} = V_{cn}\sqrt{3} < \theta_{vcn} + 30 = 100\sqrt{3} < 120 + 30 = 100\sqrt{3} < 150$$

4) Calculate the line currents

$$\text{because } \frac{van}{za} = \frac{vbn}{zb}, \text{ we can just use } I_{aA}$$

$$I_{aA} = \frac{V_{an}}{Z_a} = \frac{100vrms < 0}{100\Omega < 0} = 1 < 0 \text{ } Arms$$

$$I_{bB} = I_{aA} < \theta_{vbn} = 1 < -120$$

$$I_{cC} = I_{aA} < \theta_{vcn} = 1 < 120$$

$$I_N = |I_A + I_B + I_C| = \sqrt{I_A^2 + I_B^2 + I_C^2}$$

5) Calculate the line currents

$$I_{AB} = \frac{V_{AB}}{Z_a} = \frac{100\sqrt{3} < 30}{100\Omega < 0} = 1\sqrt{3} < 30 \text{ } Arms$$

$$I_{BC} = \frac{V_{BC}}{z_b} = \frac{100\sqrt{3} \angle -90}{100\Omega \angle 0} = 1\sqrt{3} \angle -90^\circ \text{ Arms}$$

$$I_{CA} = \frac{V_{CA}}{z_c} = \frac{100\sqrt{3} \angle 150}{100\Omega \angle 0} = 1\sqrt{3} \angle 150^\circ \text{ Arms}$$

6) Check step

For wye... $V_{ab} > V_a$ because line to line voltage must be smaller then phase voltage

For delta... $I_a > I_{ab}$ because line current must be smaller then phase current

We can also check our value for I_A

$$I_A + I_{CA} = I_{AB}, \text{ so } I_A = I_{AB} - I_{CA}, \text{ or } I_A = (\sqrt{3} \angle 30) - (\sqrt{3} \angle 150), \text{ this is in}$$

polar so need to convert to rectangular, so

$$(\sqrt{3}\cos(30), \sqrt{3}\sin(30)j) - (\sqrt{3}\cos(150), \sqrt{3}\sin(150)j) = (1.5, 0.866j) - (-1.5, 0.866j) = 3, 0j$$

Convert to polar again, $3 \angle 0$

*notes: removing the connection between neutral and common yields almost the same result for both balanced wye and delta.

It is also relevant to understand how to work with the currents. An example is shown below:

If the currents are unbalanced, draw a vector/ phasor diagram to show how the vector math $I_n = I_a + I_b + I_c$ would work, and estimate the value of the vector sum.

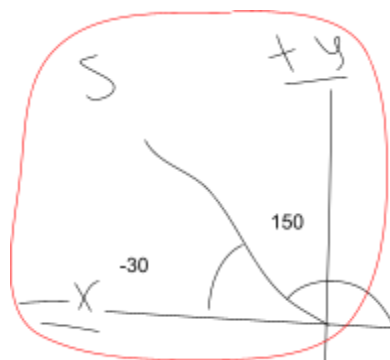
$$I_A = 1 \angle 0^\circ, \quad I_B = 2 \angle -120^\circ, \quad I_C = 3 \angle 120^\circ$$

$$I_N = 1 \angle 0^\circ + 2 \angle -120^\circ + 3 \angle 120^\circ$$

$$I_N = (1\cos 0^\circ, 1\sin 0^\circ) + (2\cos(-120^\circ), 2\sin(-120^\circ)) + (3\cos(120^\circ), 3\sin(120^\circ))$$

$$I_N = (1, 0j) + (-1, -\sqrt{3}) + (-1.5, 2.598j) = -1.5, \frac{\sqrt{3}}{2}j$$

$$I_N = \sqrt{(-1.5)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}, \angle \theta = \tan^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{-1.5}\right) = \sqrt{3} \angle 150^\circ$$



$\tan = -0.5773$ with $-x + y$, so the angle must be in quadrant 2. This makes sense because the angle is negative and $\tan$ is negative in quadrant 2

A

Positive $\tan$

T

Positive $\tan$

C

$\tan = -0.57735$, $= \tan^{-1}(-0.57737) = -30$, but when we adjust for the coordinates $= 150$
Check: $\tan(150) = -0.577$

